

PHOTONICS Research

Towards an ultrafast 3D imaging scanning LiDAR system: a review

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Light detection and ranging (LiDAR), as a hot imaging technology in both industry and academia, has undergone rapid innovation and evolution. The current mainstream direction is towards system miniaturization and integration. There are many metrics that can be used to evaluate the performance of a LiDAR system, such as lateral resolution, ranging accuracy, stability, size, and price. Until recently, with the continuous enrichment of LiDAR application scenarios, the pursuit of imaging speed has attracted tremendous research interest. Particularly, for autonomous vehicles running on motorways or industrial automation applications, the imaging speed of LiDAR systems is a critical bottleneck. In this review, we will focus on discussing the upper speed limit of the LiDAR system. Based on the working mechanism, the limitation of optical parts on the maximum imaging speed is analyzed. The beam scanner has the greatest impact on imaging speed. We provide the working principle of current popular beam scanners used in LiDAR systems and summarize the main constraints on the scanning speed. Especially, we highlight the spectral scanning LiDAR as a new paradigm of ultrafast LiDAR. Additionally, to further improve the imaging speed, we then review the parallel detection methods, which include multiple-detector schemes and multiplexing technologies. Furthermore, we summarize the LiDAR systems with the fastest point acquisition rate reported nowadays. In the outlook, we address the current technical challenges for ultrafast LiDAR systems from different aspects and give a brief analysis of the feasibility of different approaches. © 2024 Chinese Laser Press

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1. INTRODUCTION

As a kind of three-dimensional sensor, light detection and ranging (LiDAR) has emerged as one of the most eye-catching scientific research and industrial application fields of optoelectronic systems, due to its unique capabilities of long-distance and high-precision ranging [1]. This remarkable performance relies on the time-of-flight (ToF) ranging technique. Since its invention in 1961 [2], the applications of LiDAR systems have evolved over time, expanding from early uses in earth and ecological sciences [3], including remote sensing [4], topographic modeling of the moon [5], landside investigation [6], and density and temperature profiles [7], to industry [8–10] and commercial electronics [11,12]. Acting as the eyes for machines, LiDAR has gradually permeated into the daily life of people. Currently, the most prominent applications include self-driving cars [13], reconnaissance unmanned aerial vehicles [14], and

robot navigation [15]. In addition to its primary application in 3D sensing, LiDAR technology also holds promise for various contactless applications similar to radar technology [16,17], including applications in medical and health monitoring. Despite its widespread adoption in both consumer electronics and industrial domains, LiDAR systems still face several challenges. These include high device cost, low imaging resolution, and bulky size, which hinders further applications of LiDAR. Hence, technologies from different optoelectronic research fields are exploited to improve the system performance of LiDAR.

There are several metrics to evaluate the performance of a LiDAR system. The first metric is the ranging precision. LiDAR essentially obtains range information by measuring the time-of-flight of light pulses or modulated signals between the LiDAR and the target. Based on the modulation format of the light signal, current LiDAR can be categorized into three

typical types: pulsed ToF, amplitude-modulated continuous-wave (AMCW), and frequency-modulated continuous-wave (FMCW). Considering factors for different applications, such as different detection distances, system cost, and complexity, appropriate ranging methods can be chosen to enrich the applicable scenarios of LiDAR systems. For long-range detection with moderate ranging precision, pulsed LiDAR is an effective option. However, it is important to note that nanosecond pulse measurement increases the bandwidth requirement on both the receiver and the electrical circuits. AMCW LiDAR performs short-distance detection with moderate ranging precision, making it popular for indoor 3D sensing applications due to its simplicity, low cost, and compatibility with current large-scale CMOS sensors. FMCW LiDAR, based on the coherent ranging principle, is suitable for long-distance and high-precision ranging. Additionally, its interferometric scheme reduces the requirement of high bandwidth and high transmitting power. However, FMCW LiDAR involves additional components that are not involved in pulsed LiDAR and AMCW LiDAR, such as a local oscillator (LO), auxiliary interferometer, and high-precision tunable laser source, leading to increased cost and complexity.

The second metric for LiDAR systems is the stability of beam steering and the field of view (FoV) of the illuminating beam, primarily determined by the beam scanner. Traditional mechanical steering devices offer a significant advantage in scanning FoV. For example, rotating the sensor body around a motor axis leads to a 360° scanning FoV [18]. Such a system structure has been widely applied in commercial products [19]. However, stability issues arise due to rotating the entire system. Alternative beam-steering devices such as the Risley prisms [20] and microelectromechanical systems (MEMS) [21] have been proposed and adopted in LiDAR systems. Recently, solid-state beam scanners have gained much attention as a promising technology to improve the scanning performance, with advantages in robustness, high integration, and long lifetime. Various technologies have been introduced and investigated for LiDAR systems, such as an optical phased array (OPA) [22], slow light waveguide [23], focal plane switch array (FPSA) [24], and spatial light modulator [25]. However, these technical approaches also face challenges, such as the difficulty of achieving two-dimensional scanning, waveguide crosstalk, and low efficiency.

Other critical performance metrics of LiDAR systems include anti-interference ability and lateral resolution. The technologies applied in the LiDAR system continue to flourish, each with its strengths and weaknesses. This diversity makes LiDAR technology an intriguing area of research. Since the LiDAR system usually works for detecting dynamic and moving targets, the imaging rate is a significant performance requirement. The rapid acquisition time of the LiDAR system enables a high refresh rate, allowing for the capture of dynamic target information and reserving enough time for data processing, such as user tracking and decision making. For instance, on highways, fast-performing LiDAR systems are necessary to ensure adequate decision time for autonomous vehicles to safely avoid traffic accidents.

In this paper, we discuss the imaging speed of current LiDAR systems, encompassing both commercial and laboratory systems.

In Section 2, we briefly discuss the working principle of the LiDAR system and analyze the physical limitations affecting the maximum point acquisition rate. In the following Section 3, we summarize the current state of LiDAR beam scanners and elaborate on the intrinsic limitations of these devices on the 3D imaging speed. We also provide an overview of the drawbacks and strengths of these technologies. In Section 4, we mainly discuss the spectral scanning method for the LiDAR system, which can achieve an ultrafast point acquisition rate. In Section 5, we cover the parallel detection technology used in LiDAR systems, such as coding methods and multiple-detector methods, which improve the point acquisition rate. Different from a scanning LiDAR system, flash LiDAR utilizes wide-field illumination and relies on the detection array for imaging, which works similar to a camera. In Section 6, we also generalize the 3D imaging speed performance of this technology. Finally, we provide an outlook of these emerging approaches, highlighting the future applications of spectral scanning LiDAR, and then discuss the challenges that must be addressed to achieve ultrafast 3D imaging LiDAR systems. An overview of the content can also be seen as illustrated in Fig. 1.

2. SERIAL DETECTION METHOD FOR 3D IMAGING LIDAR

Compared with radar systems, LiDAR systems utilize laser beams for detection, offering high spatial resolution due to the short working wavelength of the laser source. Based on the LiDAR system scheme, there are two main types: scanning LiDAR and flash LiDAR. Scanning LiDAR is the most popular LiDAR scheme used in long-range applications, while flash LiDAR is more suitable for indoor applications with low background noise. In this section, we focus on evaluating the 3D imaging speed of scanning LiDAR systems.

Figure 2(a) illustrates the basic architecture of scanning LiDAR, which consists of three parts. The first part is a transmitter, which generates a laser beam with specific modulation, such as pulses or optical chirps. The next step involves deflecting the probe beams towards different positions on targets, which is crucial for obtaining 3D information. Optimizing the laser beam with a narrow divergent angle ensures high lateral resolution. The beam-steering device is responsible for achieving dense detecting points, contributing to the superior angle resolution of LiDAR systems when compared with other ranging techniques. In the receiving end, an optoelectronic converter captures the optical signal. Based on the requirement of the system receiving sensitivity or ranging function, various receivers can be used, such as avalanched photodiodes [35], silicon photomultipliers [36], balanced photodetectors [37], or single-photon photodetectors [38,39]. As depicted in Fig. 2(a), the LiDAR system operates in serial detection mode. The transmitter launches the first probe beam towards the surface of the targets, where the reflection scattering occurs. Once the receiver captures the echo signal, the depth information of the first position can be obtained. The beam-steering device then directs the second probe beam towards other points on the target, and this process repeats for 3D scanning. Figure 2(b) illustrates this process, which involves three ranging techniques working in serial detection mode. Here, we use repetition rate to define

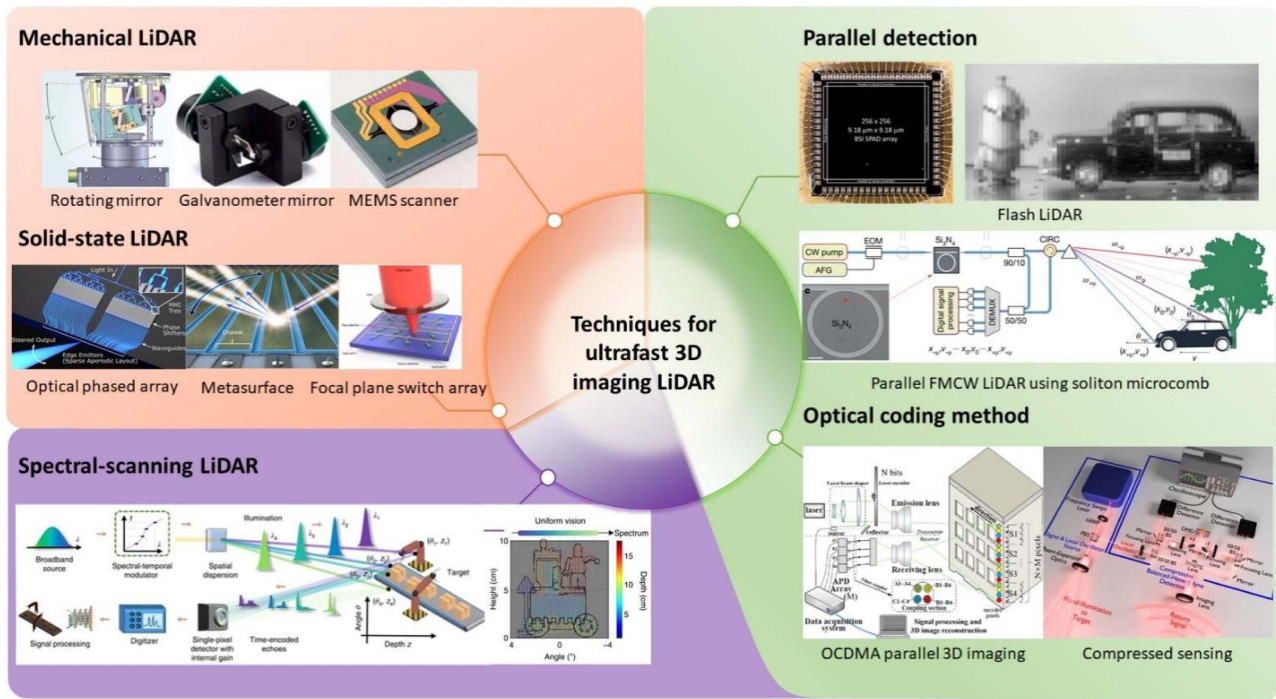


Fig. 1. Overview of 3D imaging LiDAR. Rotating mirror reprinted with permission from Ref. [13], copyright 2010, Springer Nature. Galvanometer mirror reprinted with permission from Ref. [26], under a Creative Commons Attribution 4.0 International License. MEMs mirror reprinted with permission from Ref. [27], copyright 2021, Optica Publishing Group. Optical phased array reprinted with permission from Ref. [28], copyright 2020, Optica Publishing Group. Metasurface reprinted with permission from Ref. [25], copyright 2020, Springer Nature. Focal plane switch array reprinted with permission from Ref. [29], copyright 2022, Springer Nature. Spectral scanning LiDAR reprinted with permission from Ref. [30], copyright 2020, Springer Nature. Flash LiDAR reprinted with permission from Ref. [31], copyright 2019, IEEE. Parallel FMCW LiDAR using soliton microcomb reprinted with permission from Ref. [32], copyright 2020, Springer Nature. OCDMA parallel 3D imaging reprinted with permission from Ref. [33], copyright 2020, Optica Publishing Group. Compressed sensing reprinted with permission from Ref. [34], copyright 2019, Optica Publishing Group.

the frequency of launching the probe beam, which is $1/T$ as depicted in Fig. 2(b). The pulsed ToF method measures the distance by calculating the round-trip delay time τ , while AMCW ToF and FMCW LiDAR make use of modulated optical signal rather than pulsed optical signal for sensing. However, all these methods with single-pixel detection require launching the next probe beam signal after receiving the echo signal, as shown in Fig. 2(c). This procedure poses a physical limitation on the maximum point acquisition rate. The repetition rate, also referred to as the shot rate, is equivalent to the point acquisition rate. Thus, the maximum repetition rate determines the point acquisition rate.

Here, we discuss factors affecting the maximum point acquisition rate under quantitative analysis. We focus on the speed limitations of the optical system, excluding the time consumption on electronic circuits. Assuming a detection range of over 1.5 m, the round-trip time delay τ can be estimated to be more than 10 ns. In pulsed LiDAR, the impact of pulse width on the point acquisition rate can be neglected compared to τ , as the pulse width is typically within several nanoseconds. As shown in Fig. 2(c), the repetition rate $1/T$ relies on the maximum detectable range, which is equal to $c\tau/2$. Specifically, for a scene with a maximum measuring distance of 150 m, the repetition rate should be lower than 1 MHz, indicating a

maximum point acquisition rate of 1 MHz. If the repetition rate exceeds this value, the measurement distance corresponding to each point could be inaccurate, leading to range ambiguities. Consequently, for pulsed LiDAR, the physical limitation on the maximum point acquisition rate is determined by the time-of-flight delay τ .

For AMCW and FMCW methods, we observe that the shot rates of laser sources could not match the speed of pulsed LiDAR. This is due to the longer modulation time compared to the time-of-delay τ , as depicted in Fig. 2(c). In the case of FMCW LiDAR, the echo signal must temporally overlap with the local signal to generate a beat signal. For AMCW LiDAR, the phase difference between the echo and local signals should be smaller than 2π . But the maximum point acquisition rates of both systems and the acquisition rate determined by the maximum measurement distance are within the same order of magnitude. In summary, the round-trip time delay τ limits the maximum point acquisition rate of scanning LiDAR systems. Compared with pulsed LiDAR, the maximum point acquisition rates of scanning FMCW and AMCW LiDAR systems are lower due to the modulation time.

The above discussion is based on theoretical analysis. However, it is essential to note that the response time of beam-steering devices significantly influences the point

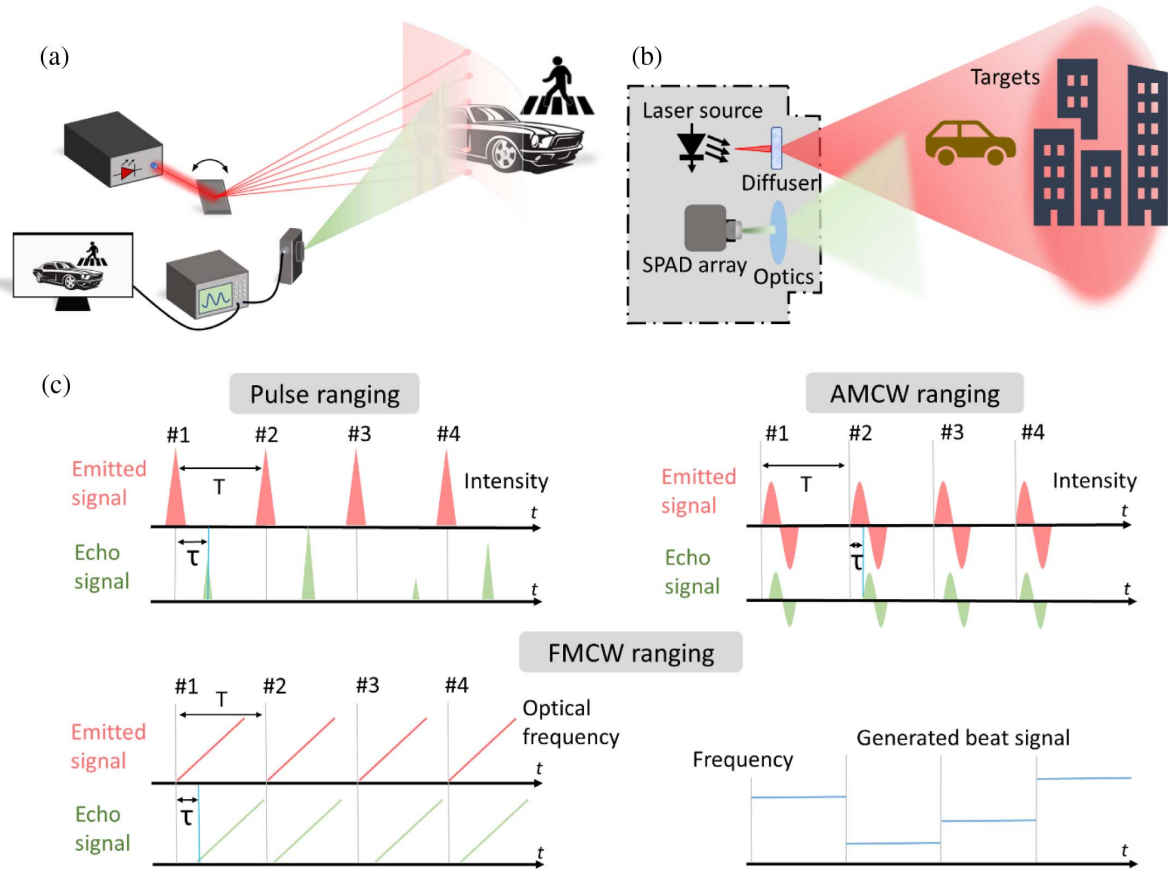


Fig. 2. (a) Basic architecture of a serial detection 3D imaging LiDAR system. The type of light source used in the LiDAR system depends on the ranging technology employed. For pulsed LiDAR, laser diodes or mode-locked fiber lasers can serve as the light source. For AMCW or FMCW LiDAR, laser diodes or other semiconductor lasers capable of transmitting continuous waves can be used. Here modulation devices are employed to generate different signals, although they are not all depicted here. (b) Basic architecture of parallel detection LiDAR, flash LiDAR, for instance. (c) Working principles of different ranging technologies for the serial detection method.

acquisition rate. At present, the beam-steering rate of most beam scanning devices is far lower than the upper rate limit determined by the round-trip delay time, except for spectral scanning LiDAR. In the following section, we will introduce different types of beam scanners used in LiDAR systems and discuss their performance in terms of beam-steering speed.

3. BEAM-STEERING DEVICES FOR LIDAR SYSTEM AND THEIR SCANNING SPEED

As discussed in the previous section, current beam-steering devices used in the LiDAR system limit the point acquisition rate. In this section, we summarize the beam-steering methods implemented in LiDAR systems and analyze the main factors influencing the beam-steering speed. Based on the beam-steering mechanism, we categorize beam-steering devices into two groups: mechanical beam scanners and solid-state beam scanners. We introduce and analyze beam scanners in ascending order of scanning rate, from slow to fast. Here, the spectral scanning method will be separately discussed in the next section, due to the high performance of such a technique in beam-steering speed. The point acquisition rate is used to represent

the scanning speed of beam scanners. Throughout this introduction, unless specifically mentioned, the data point acquisition rate of a given scanner is estimated based on the use of a single laser source and receiver for detection.

A. Mechanical Beam Scanners

Mechanical beam scanners are currently prevalently utilized in commercial LiDAR systems. As the name indicates, these scanners employ a mechanical system to deflect laser beams by vibrating a mirror driven by a motor. Benefiting from the high rotational capability of the motor, mechanical scanners can achieve a 360° scanning FoV. However, mechanical scanners are typically limited to one-dimensional beam steering, requiring the use of multiple light sources and detectors for parallel detection to achieve beam steering in the second direction. Thus, in the subsequent discussion regarding the scanning speed performance of various mechanical scanners, we specifically focus on the beam-steering speed of a single laser beam. We use line scan rate to evaluate the beam-steering performance of the mechanical scanners, which represents the reciprocal of the time required for a complete sweep along one axis. The works from industry and academia are included.

1. Polygonal Scanners

A polygonal mirror refers to the scanner with multiple reflective facets. Typically, a metal mirror is used for beam deflection. Polygonal mirrors offer a relatively high scan rate compared to other mechanical beam scanners. Different types of polygonal scanners are illustrated in Fig. 3(a) [40,42]. By rotating the central axis, beam steering can be achieved. The scanning pattern produced by a polygon mirror is rectangular with raster lines. The scanning speed of polygonal mirror is determined by the rotating speed of the bearing system. Turbine drives, in comparison to electronic motors, can achieve fast acceleration and high speeds, albeit at a higher cost and shorter lifespan. The rotating speed of such drives can reach the MHz level. The maximum point acquisition rate of a standard polygon mirror can reach up to 100 kHz [43]. One limitation of the polygon mirror is limited FoV. Thus, a polygon mirror is mostly suitable for fast scanning illumination in small regions, while off-axis parabolic mirrors with large FoV can be applied in full-field scanning [44].

2. Galvanometric Mirrors

The galvanometer scanner consists of a rotating mirror and a moving magnet torque motor. The motor operates to achieve beam deflection. The design of the motor incorporates a permanent magnet. The overall architecture and the detailed structure of the galvanometer-based scanning motor are shown in Fig. 3(b). A high-quality galvanometer scanner exhibits a robust linear relationship among torque, current, and angular position. The response time for a small angle step decides the maximum beam-steering speed of this device. Usually, a mobile magnet is used over a stationary magnet and a rotating coil to achieve the fast response time and high resonant frequency of the system. The response time is at the level of 100 μ s, resulting in a line scan rate of approximately 100 Hz and a point acquisition rate of around 10 kHz [45,46].

The scanning speed of a galvanometric scanner is not fast and its field of view is limited due to the size of the mirror. Although it is realizable to achieve 2D beam steering by cascading two scanners working in different axis directions

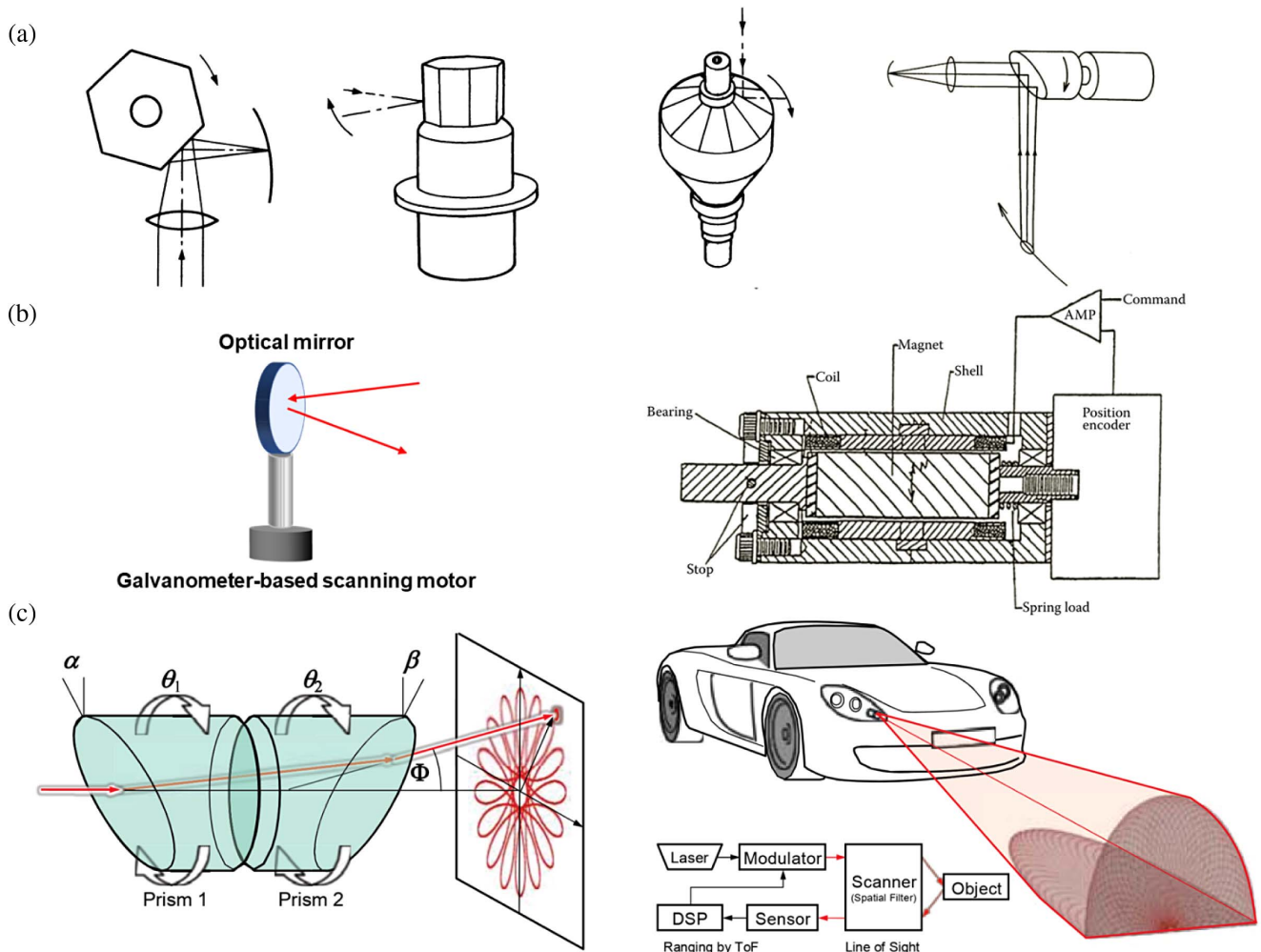


Fig. 3. Mechanical beam scanners. (a) Typical polygonal prism configurations: convergent beam scanning, regular polygonal scanner, pyramidal mirror scanner, and single-faceted cantilevered scanner. (b) Left: basic structure of galvanometric scanner. Right: structure of a magnet torque motor. (c) Left: working principle of beam steering by using a pair of prisms. Right: schematic diagram of applying Risley prism in vehicle LiDAR. (a), (b) Reprinted from Ref. [40], under a Creative Commons Attribution 4.0 International License. (c) Reprinted with permission from Ref. [41], copyright 2018, IEEE.

[26,47,48], a galvanometric scanner is not a prevalent method in LiDAR applications.

3. Risley Prism

The Risley prism scanner can achieve high-accuracy beam steering benefiting from the prism refraction principle. By rotating the Risley prism system, a large scanning FoV can be realized. As a result, the Risley prism has been employed in various fields, such as confocal microscopy [49], bioimaging [50], satellite laser ranging [51], and free space connection [52]. The basic architecture of a Risley prism for beam steering can be seen in Fig. 3(c). A Risley prism scanner is composed of two identical components, but with different wedge angles, refractive indices, and rotating speeds. One of the major challenges in this application is exploring the relationship between the scanning track and the position of prisms. The scanning pattern of prisms is highly complex and a blind zone often occurs. Extensive theoretical and experimental analyses have been conducted to investigate the relationship between the scanning patterns and the rotation control [53] and to address the issue of blind zones [20,41,54–57].

Motorized rotation stages are commonly employed to individually control the rotation of each prism. By coordinating the mechanical motion of the prisms, various beam-scanning trajectories and beam-steering speeds are obtained [58]. The beam-steering speed of Risley prism scanners largely depends on the rotation speed of the motor. To realize smooth and stable beam steering, precise control of driving motors is crucial, with high controllability and acceleration ability. Thus, further acceleration of the speed will lead to higher control complexity and cost [59]. By using a commercially available electronic motor, the point acquisition rate can range from 1 kHz to 10 kHz. By employing multiple laser sources and receivers, a commercial LiDAR system based on the Risley prism can achieve a maximum point acquisition rate of 480 kHz in total [60].

4. MEMS Scanners

Microelectromechanical systems (MEMS) scanning mirrors can be considered as a quasi-solid-state method. Not only for LiDAR systems, MEMS scanners can also be applied in optical wireless communication [61], projection displays [62,63], and medical imaging [27,64,65]. Compared to other mechanical beam scanners, MEMS scanners offer considerable advantages in size, cost, power consumption, and beam-steering speed. The high performance of MEMS scanners requires precise control of the mirror. The actuator is the main component for generating a high drive torque for beam steering. For MEMS scanners, the performance mainly depends on the actuation. There are four typical actuation methods for MEMS scanners: electrostatic (ES) actuation, electromagnetic (EM) actuation, piezoelectric (PE) actuation, and electrothermal (ET) actuation [66]. However, the ET actuation exhibits a long response time, making ET scanners unsuitable for LiDAR application [67]. ES actuation has the advantages of fabrication compatibility, easy integration, and long-term stability [68–72]. EM actuation shows a linear response for generating linear slow scanners but the scanning speed is relatively low [73–75]. PE-based MEMS scanners can be operated at low voltage and deliver high force but with short stroke length [76–78]. The basic architecture of ES [71], EM [79], PE

[80], and ET [21] actuation is shown in Figs. 4(a)–4(c). MEMS scanners based on EM and ET actuation exhibit low scanning speed. In contrast, PE and ES scanners belong to the category of fast scanners with high resonant frequency suitable for fast 3D imaging LiDAR. Based on the data summarized in Ref. [66], we list several resonant frequencies of different types of fast MEMS scanners in Table 1.

As depicted in Table 1, MEMS scanners exhibit line scan rates exceeding 10 kHz. However, considering the scanning FoV and the aperture size, the MEMS scanner LiDAR usually could not work at such high-frequency resonance. To address this limitation and improve the point acquisition rate, MEMS scanner arrays have been proposed for the parallel detection LiDAR application [81]. By adopting three quasistatic SVC comb drive MEMS scanners, a 2D raster scanning 3D ToF LiDAR is realized with 1-MHz data acquisition rate [82]. The photograph of such ToF LiDAR can be seen in Fig. 4(d). However, the whole volume is substantial, posing challenges for impact integration or lightweight commercial applications. To ensure a balance in critical technical parameters for a wide range of applications, a semisolid micromechanical beam-steering system has been proposed. This system integrates metasurfaces with a decentered microlens array within a compact size of less than 0.5 mm³, enabling diffraction-limited resolution across a large field of view measuring 30° × 30° [83]. With the commercial piezoelectric ceramic, the scanning speed has the potential to exceed 10 kHz. A comparison of LiDAR performance for selected mechanical beam scanners is presented in Table 2.

B. Solid-State Beam Scanners

Mechanical scanners, relying on motors to deflect the beam, suffer from numerous drawbacks, including bulkiness, high cost, short lifetime, instability, and low scanning rate. Therefore, solid-state beam scanners are widely studied. As the name suggests, solid-state beam scanners are characterized by the absence of moving components, which gives scanners stability and integrity. The primary objective of all the research groups in this field is to advance LiDAR technology for reductions in size and cost and enhanced stability and speed. In this subsection, we will focus on various types of solid-state beam scanners and their performance on scanning speed.

1. Optical Phased Array

Facing future integrated optical components, silicon photonics is a popular technology that enables the integration of optical elements on a single chip. An optical phased array is one of the critical developed integrated beam scanners for LiDAR systems. The complete structure of an optical phased array is depicted in Fig. 5(a). Within the OPA, several optical couplers or multi-mode interference (MMI) splitters are used to separate light signals into each antenna. Phase shifters are incorporated in each channel to modulate the phase information.

Different phase delays enable the manipulation of wavefront propagation direction, thereby enabling beam-steering functionality. Phase shifters play a crucial role in OPAs [88], and commonly there are two effective methods for phase tuning: thermo-optic (TO) phase shifters and electro-optic (EO) phase shifters. The structures of a TO phase shifter and EO shifter are shown in Fig. 5(b) [84]. TO phase shifters utilize the

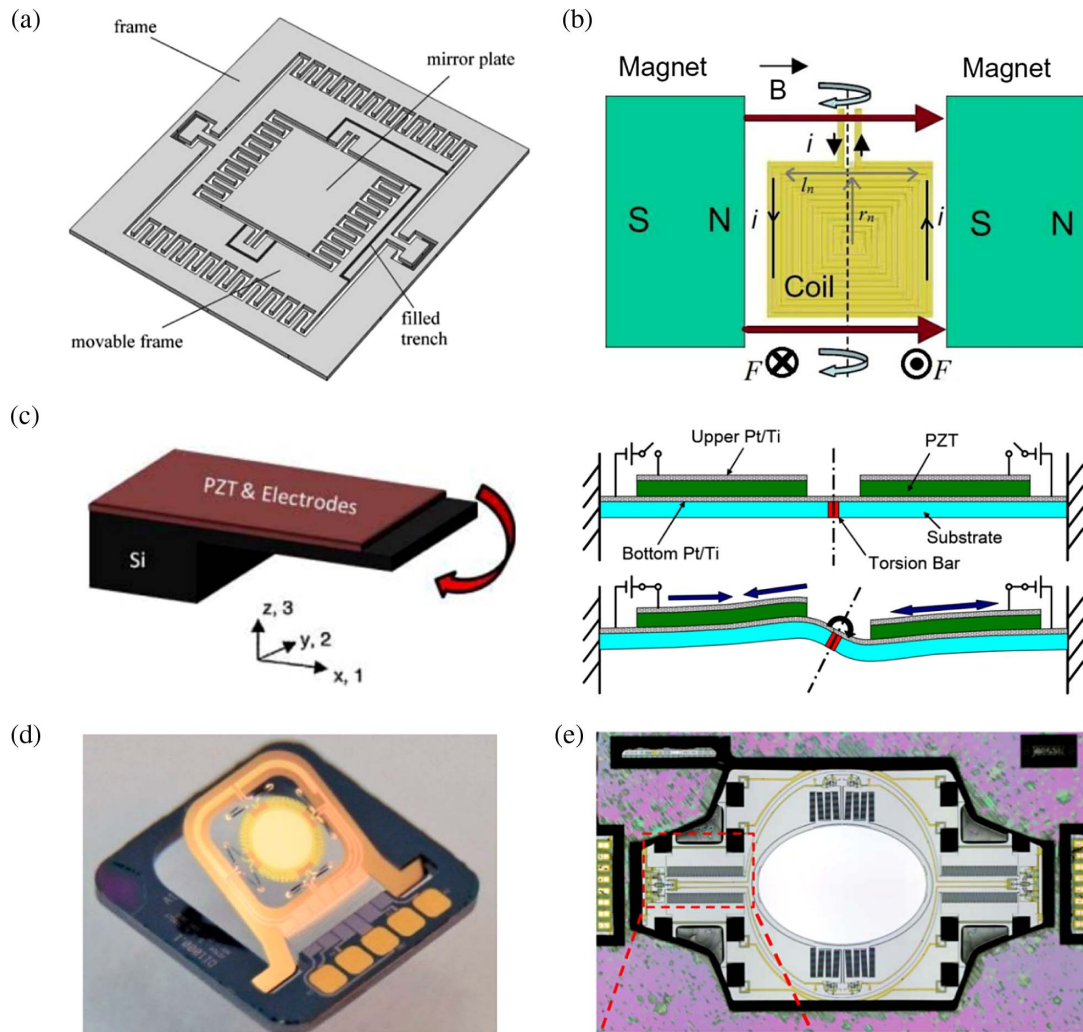


Fig. 4. MEMS beam scanners. (a) 2D MEMS scanner based on ES actuation. (b) EM MEMS scanner structure with coils and magnets. (c) Basic working principle and schematic of PE MEMS scanner. (d) ET MEMS scanner. (e) Fast MEMS scanner with quasistatic resonant actuation. (a) Reprinted with permission from Ref. [71], copyright 2001, Elsevier. (b), (c), (e) Reprinted from Ref. [66], copyright 2014, IEEE. (d) Reprinted from Ref. [21], under a Creative Commons Attribution 4.0 International License.

Table 1. Simple Summary of MEMS Scanners with High Resonant Frequency

Actuation	Resonant Frequency (kHz)	Ref.
ES	30.8	[69]
ES	30.8	[70]
EM	21.3	[73]
EM	21	[74]
PE	40	[75]
PE	40	[78]

thermo-optic effect to change the phase information by altering the refractive index of materials through temperature variation. By depositing a metal layer on the upper cladding, the temperature of waveguides can be changed due to the Joule effect. However, the TO phase shifters suffer from low operating speed. Thus, an OPA based on thermo-optic phase shifters is unsuitable for fast beam steering.

In contrast, electro-optic phase shifters modulate the refractive index by controlling the injecting charge. This allows for high-speed beam steering due to the fast tuning speed of electronic signal switches. However, EO shifters exhibit higher insertion loss compared to TO shifters. In Ref. [89], by using a silicon thermo-optic phase shifter, an optical phased array was demonstrated with a scanning rate in the tens of kHz range. For thermo-optic phase shifters, the scanning rate is challenging to reach megahertz level [90]. Researchers have made advancements in improving the scanning rate of OPA. For instance, an electrostatically actuated linear phase modulator with a high refresh rate was utilized, resulting in an improved scanning rate of 350 kHz [91]. Gozzard *et al.* presented an OPA with high-bandwidth EO shifters, enabling arbitrary control of the wavefront and realizing MHz beam-steering speed [92]. A 1×16 silicon phased array using electro-optic phase shifters attains an operation speed of 20 MHz [93]. Based on GaAs/AlGaAs multiple quantum well phase modulators, an OPA

Table 2. Comparison of LiDAR Performance for Selected Mechanical Beam Scanners

Ref. and Type	Mechanical Scanners	FoV (°)	Spatial Resolution (°)	Mirror Diameter (mm)	Line Scan Rate (Hz)	Point Acquisition Rate (kHz)
[43] Commercial	Polygonal mirror	—	—	76.2	30,000	~100
[46] Commercial	Galvanometric mirrors	40	—	3–50	~100	~10
[60] Commercial	Risley prism	70.4×77.2	0.23	61.2	—	480
[82] Research	MEMS mirrors	60×40	—	3.6	—	1000

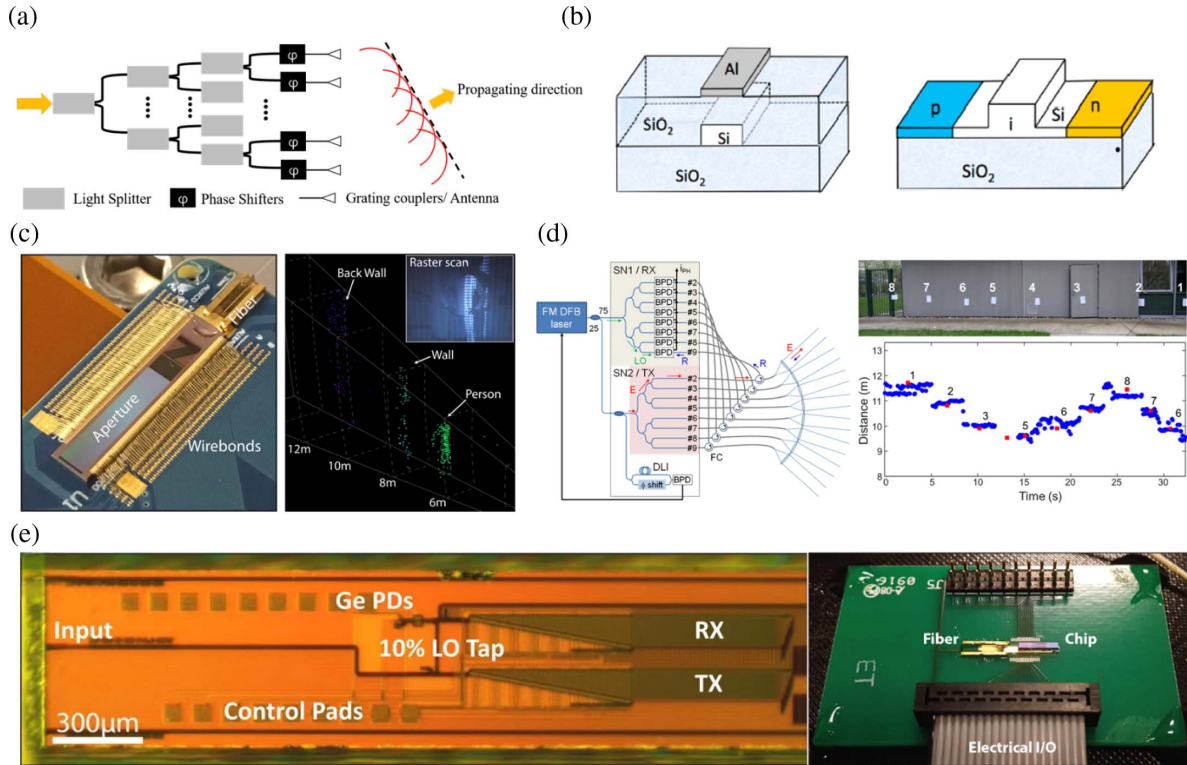


Fig. 5. Optical phased array. (a) Working principle of optical phased array (OPA). (b) Schematic diagram of TO and EO phase shifters. Heater metal layer is made of aluminum or other metals. (c) Picture of 512-element optical phased array with an inline architecture and the 3D imaging results. (d) Architecture of FMCW LiDAR system for long range detection and its one-dimensional scanning ranging results. (e) OPA device and the packaged system with epoxied fiber of the first OPA-based coherent LiDAR system from the group at MIT. (b) Reprinted with permission from Ref. [84], copyright 2021, IEEE. (c) Reprinted with permission from Ref. [85], copyright 2019, IEEE. (d) Reprinted with permission from Ref. [86], copyright 2018, IEEE. (e) Reprinted with permission from Ref. [87], copyright 2017, Optica Publishing Group.

with even GHz-level deflection speed was reported [94]. Similarly, LiNbO₃-based phase modulators can also realize similar high-speed capabilities [95]. In summary, OPAs exhibit significant potential for future applications facing fast and integrated 3D imaging LiDAR, with point scanning rates even reaching GHz level. By now, the application of OPAs in LiDAR systems is still developing. As shown in Figs. 5(c)–5(e), three typical application examples are demonstrated. The first integrated coherent LiDAR based on chip-scale OPA was achieved in 2017 by the research group from Massachusetts Institute of Technology [87]. Long-range detection by 1D and 2D OPAs was also demonstrated [85,86]. A novel chip-based optical phased array operated at visible wavelength has been designed [28]. The device can greatly reduce the size of light projection components used for augmented reality and various applications.

2. Focal Plane Switch Array

Focal plane switch arrays are commonly referred to as lens-assisted beam scanners [96]. Recently, this approach has emerged as one of the hottest research topics with chip-scale beam scanners in LiDAR systems. FPSA operates similar to a camera, utilizing a lens to establish a one-to-one correspondence between the output direction of a laser beam and position of each antenna. The working principle of FPSA for beam steering can be observed in Fig. 6(a).

Here each antenna directs a laser beam into free space at a specific position with the switch array located at the focal plane of a lens. Similar to OPA, FPSA enables robust solid-state beam steering with a compact size and low cost. In addition, both techniques are primarily implemented on the silicon photonics platform, which introduces insertion loss to some extent. Regarding OPA, it is still a bottleneck to achieve large-scale

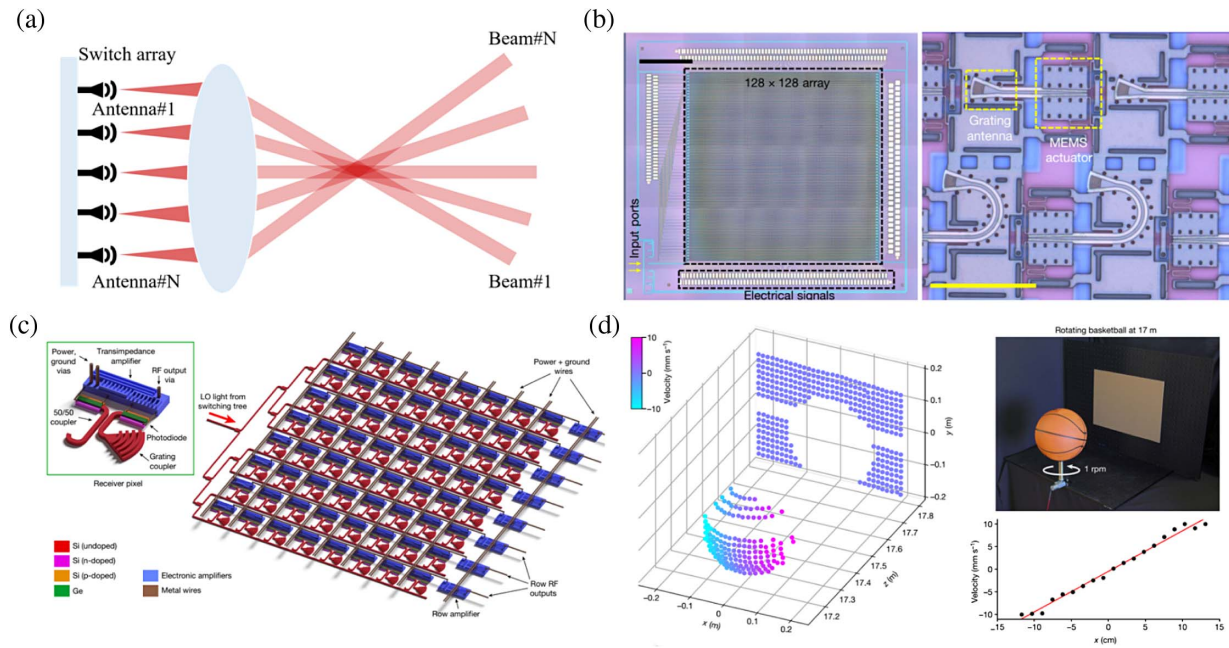


Fig. 6. Focal plane switch array. (a) Working principle of FPSA for beam steering. (b) Microscopic images of a large-scale FPSA beam scanner. (c) Schematic diagram of an integrated FPSA chip and (d) velocity measurements by the FPSA LiDAR. (b) Reprinted with permission from Ref. [29], copyright 2022, Springer Nature. (c), (d) Reprinted with permission from Ref. [97], copyright 2021, Springer Nature.

2D integration. Although some works have been reported for 2D OPA, it is still inadequate for LiDAR applications [25,98]. On the other hand, it is feasible for FPSA structure to integrate a large number of antennas on a chip, since each antenna is independent of the others. In the case of FPSA scanners, the performance of an optical switch is crucial. Cao *et al.* successfully demonstrated an FPSA with 16 pixels using thermally tuned Mach–Zehnder interferometer (MZI) switches [99], but the integration density is limited by the large footprint of the MZIs and the high power consumption of thermo-optic phase shifters. In other works, MHz-level scanning rates were realized by using Mach–Zehnder switches [24,100,101]. A microelectromechanical silicon photonic switch is another method to control the optical antennas. As depicted in Fig. 6(b), Ming C. Wu's group from UC Berkeley reported a breakthrough in integrated LiDAR with a 128×128 -element FPSA featuring a wide field of view of $70^\circ \times 70^\circ$ [29]. The MEMS-based FPSA has a microsecond response time, corresponding to MHz-level scanning rate. The entire size of the scanner chip, measuring $10 \text{ mm} \times 11 \text{ mm}$, signifies an advancement in integrated LiDAR technology. By increasing the chip size or shrinking the footprint of each pixel, the shortcoming of low lateral resolution can be solved [102]. A MHz mechanical resonant frequency beam scanner has also been reported by using a novel polysilicon grating switch [103].

An ultra-compact integrated FPSA LiDAR system was proposed by Rogers *et al.*, which represents a significant step towards future integrated FPSA LiDAR systems [97]. The basic diagram of their proposed chip is depicted in Fig. 6(c). In this study, a large-scale coherent FPSA scheme was demonstrated on the chip scaled to $6 \text{ mm} \times 3 \text{ mm}$. By using a 512-pixel

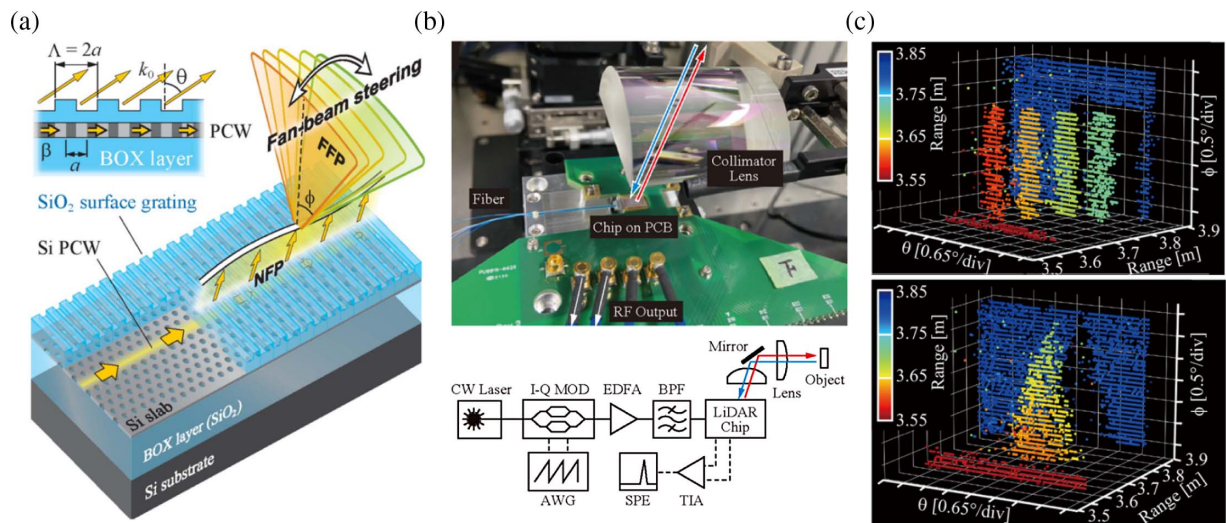
coherent detector array, the system achieved ranging at 75 m with an accuracy of 3.3 mm.

Since FPSA-based beam scanners have developed in the last few years, there is still limited research in this area. The scanning speed of such method can be greatly improved by using switches with fast response time, potentially reaching scanning rates at the GHz level, similar to the capabilities of OPA systems.

3. Slow-Light Grating Beam Scanner

The study of beam scanners with photonic crystal slow-light gratings fabricated with Si photonics has also been explored for LiDAR applications. Based on the working principle, slow-light grating beam scanners can be considered as a special type of FPSA.

The basic structure and working principle of a slow-light grating beam scanner are shown in Fig. 7(a) [104]. Light is incident into a photonic crystal waveguide and is then emitted to free space through a surface grating. By tuning the wavelength, the output beam direction can be changed correspondingly. A prism lens could be used for beam collimation and to suppress the beam divergence [106]. By switching the photonic crystal waveguide, beam steering of the other dimension is also achieved and thus a 2D beam-steering function is realized. Previous studies equipped a thermo-optic shifter as the optical switch, limiting the beam-steering rate to 100 kHz [107,108]. The first FMCW LiDAR chip using a slow-light grating beam scanner was presented by Toshihiko Baba's group [105]. The picture and schematic diagram of the LiDAR system are depicted in Fig. 7(b). The beam switching time is $2.7 \mu\text{s}$, allowing for MHz scanning rate. The 3D imaging performance can be seen in Fig. 7(c). The slow-light-scanner-based LiDAR system



presents an alternative approach to OPA and FPSA for on-chip LiDAR systems.

4. Metasurface-Based Beam Scanner

With the rapid development of nanophotonics, optical beam scanners and flat optical devices based on metasurfaces have gained attention for LiDAR application. To dynamically adjust a metasurface for beam steering, one approach involves changing the physical structure of the metasurface (e.g., MEMS) or elastic deformation of the materials. Another method involves tuning the optical properties of the materials utilized in the construction of a metasurface [109]. By leveraging subwavelength phase control with nanostructure elements, metasurface-based beam scanners can achieve precise wavefront deflection [110]. These scanners rely on the active medium to perform a beam-steering function, with commonly used materials including transparent conducting oxide (TCO), multiple quantum wells (MQWs), and liquid crystals [110]. A TCO-based beam scanner achieves a high-speed deflection rate (up to 10 MHz) by applying a low voltage bias [111]. MQW-based beam scanners can work in the visible and shorter near-infrared bands. By using high-speed electrical modulation, the beam-steering rate is promising to reach gigahertz speeds [112]. Liquid crystal spatial light modulators can change the wavefront by modulating the phase information of the laser beam with liquid crystals. The scanning rate is relatively low, typically in the kilohertz range, due to the limitation of the low modulation speed of liquid crystal [113]. The basic structures of these three kinds of metasurface-based beam scanners are shown in Figs. 8(a)–8(c).

The work published by Samsung has successfully demonstrated the feasibility of using solid-state active metasurfaces for LiDAR applications [25]. The structure of the spatial light modulator is shown in Figs. 8(d) and 8(e). The all-solid-state metasurface array enables continuous beam steering over a range of 0° to 360° , with a scanning rate of 5.4 MHz. In their

experimental results, they successfully measured targets placed at the distance of 4.7 m, validating the potential application of SLM in LiDAR systems. However, the challenge of low diffractive efficiency remains to be addressed.

5. SLM-Based Beam Scanner

The spatial light modulators (SLMs) are devices used in the field of optics to control and manipulate light waves, offering precise control over the spatial distribution of light through modifying phase, intensity, or polarization. Holoeye provides SLMs that function as dynamic optical elements, enabling rapid and flexible beam steering [114]. Compared to metasurface-based beam steering, SLMs offer greater flexibility in beam shaping and steering, outperforming MEMS devices in this domain. In 2018, Jian Xu and his colleagues introduced an innovative approach combining SLM with a metasurface, resulting in doubling the range of beam steering to 160° . Furthermore, they successfully achieved an outstanding resolution of 0.017° simultaneously [115].

Wavefront shaping, a technology that involves calculating the transmission matrix in multimode optical fibers, allows for the pre-shaping in the input optical field to achieve a desired output [116]. In a proposed scanning-based MMF LiDAR system, raster scanning was realized by illuminating the proximal facet with a modulated input light field [117]. Before input light enters the MMF, a digital micromirror device (DMD) is employed for wavefront shaping and adjusting the focal point on the output side. The overall configuration of the system is illustrated in Fig. 9(a). The beam-steering speed depends on the operating speed of DMD, which was 22.7 kHz in this study. The system achieved a point acquisition rate of around 23 kHz, capable of near-video frame rates for 3D imaging of scenes. Smith *et al.* conducted an examination of the beam-steering capabilities of four distinct commercial DMDs, as shown in Fig. 9(b). They integrated these DMDs with a single-chip LiDAR, resulting in an overall system capable of live

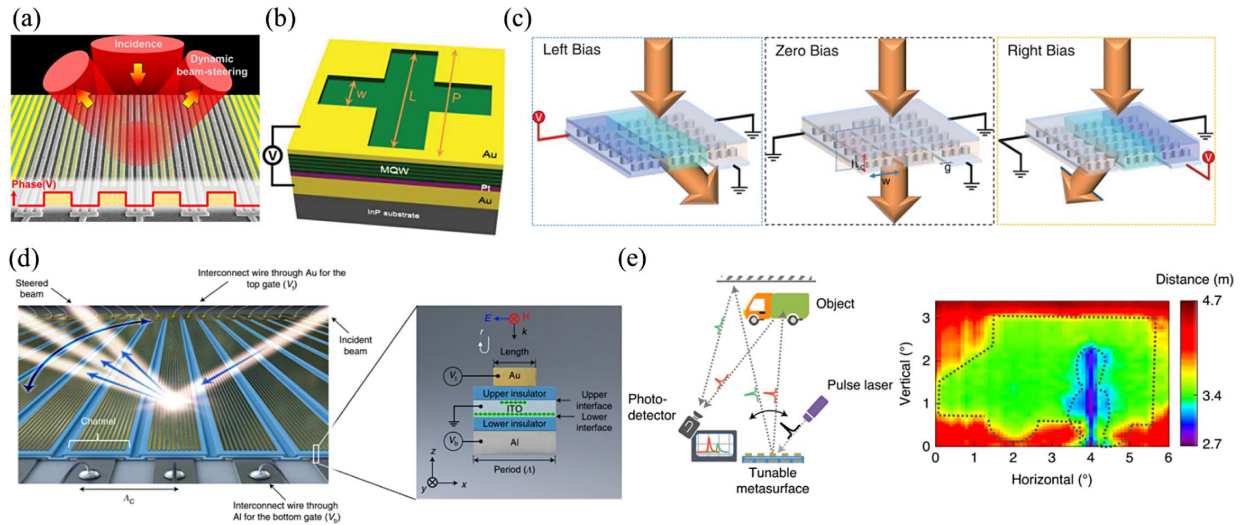


Fig. 8. Metasurface-based beam scanner. (a) Beam-steering performance operated by the TCO material approach. (b) Schematic diagram of MQW metasurface. (c) Liquid-crystal-based SLMs for beam steering. (d) Illustration of active metasurface array for beam steering. (e) Schematic diagram of the 3D LiDAR experimental setup and the 3D depth image produced using the metaphotonic SLM. (a) Reprinted with permission from Ref. [111], copyright 2016, American Chemical Society. (b) Reprinted with permission from Ref. [112], copyright 2015, Wiley-VCH. (c) Reprinted with permission from Ref. [113], copyright 2019, AAAS. (d), (e) Reprinted with permission from Ref. [25], copyright 2020, Springer Nature.

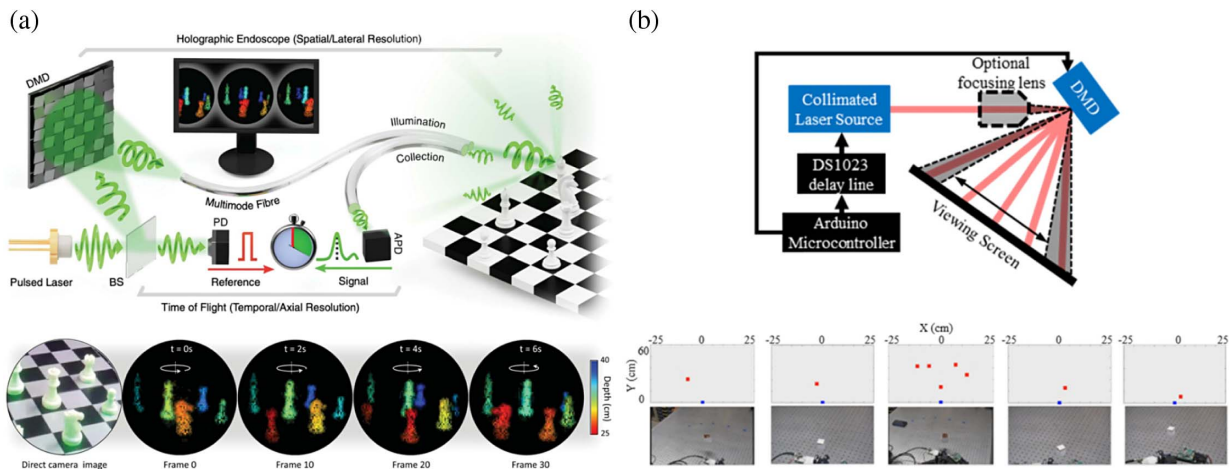


Fig. 9. DMD-based beam scanner. (a) System setup, true scene, and depth resolved images of the proposed LiDAR system. Recording of 3D imaging results at a 5 Hz frame rate. (b) Top: illustration of the optical system and beam-steering scheme. Bottom: representation of a captured movie of the LiDAR system capturing swinging pendulums placed in each of the five scanning diffraction orders. (a) Reprinted with permission from Ref. [117], copyright 2021, AAAS. (b) Reprinted with permission from Ref. [118], copyright 2017, Optica Publishing Group.

imaging at a rate of 3340 points per second and offering a full field of view of 48° [118]. This application served as a substitute for the conventional MEMS scheme, as it contributed advantages such as a larger steering angle, a larger beam size, and minimized beam divergence.

6. Acousto-Optic Beam Scanner

In addition to the aforementioned electronic-optic deflectors, acoustic-optic beam scanners have emerged as a viable option in the domain of beam scanning technology. Acoustic waves propagating through a material mechanically change its

refractive index [119], as depicted in Fig. 10(a). This periodic variation in refractive index acts like an optical grating, moving at the speed of sound in the crystal, thus diffracting a laser beam via Brillouin scattering. A two-dimensional scanner can be formed by arranging two AODs orthogonally in series [119]. A recent development involves an on-chip acousto-optic beam-steering technique that employs a single gigahertz acoustic transducer to achieve frequency-angular resolving LiDAR [120]. Capitalizing on the principles of Brillouin scattering, whereby differently steered beams carry distinct frequency shifts, this approach facilitates frequency-angular resolving

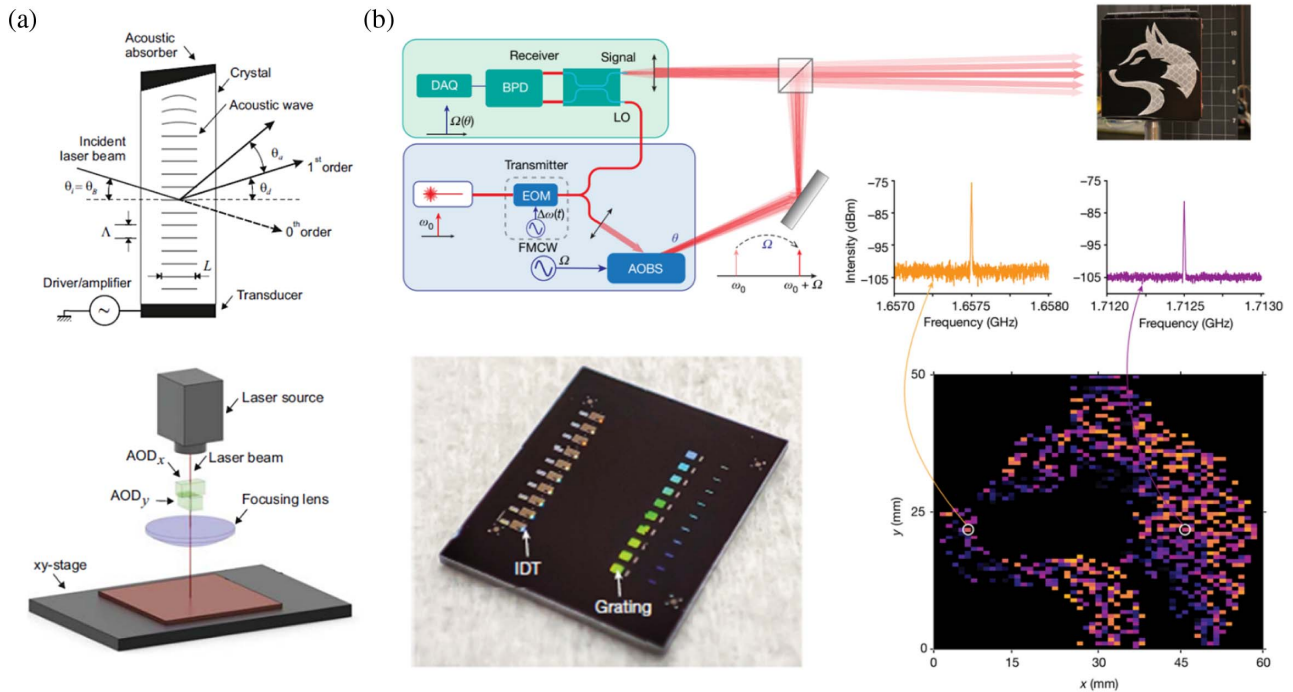


Fig. 10. (a) Top: typical configuration of an acousto-optical deflector. Bottom: two-dimensional scanner formed by arranging two OADs orthogonally in series. (b) Top: schematic of frequency-angular resolving LiDAR using acousto-optic beam scanner. Bottom left: photograph of an LNOI chip with 10 acousto-optic beam scanners. Bottom right: LiDAR imaging results and the beating signal. (a) Reprinted with permission from Ref. [119], copyright 2014, Elsevier. (b) Reprinted with permission from Ref. [120], copyright 2023, Springer Nature.

LiDAR. The system successfully performed FMCW ranging with an 18° field of view, 0.12° angular resolution, and ranging distance up to 115 m. The imaging results are shown in Fig. 10(b). The single prototype has an electronics-limited switching speed of $1.5 \mu\text{s}$, corresponding to an imaging rate of 0.67 megapixels per second with high-speed detectors. If using 16 channels for imaging, one device provides an imaging rate of more than 10 megapixels per second. The demonstration can be scaled up to an array configuration, realizing miniature, low-cost frequency-angular resolving LiDAR imaging systems with a wide 2D field of view.

4. SPECTRAL SCANNING LIDAR SYSTEM FOR ULTRA-FAST 3D IMAGING

The spectral scanning technique, utilizing diffractive components and a wavelength-swept laser source, enables inertia-free solid-state beam steering. The inherent passive beam deflection of optical diffraction facilitates the beam-steering performance in LiDAR applications. The highlight of the spectral scanning method is undoubtedly the ultra-fast scanning rate, with the reported maximum scanning rate reaching GHz level [121]. Previously, the spectral scanning method has been applied in various fields, such as optical imaging [122], photography [123], and microscopy [124] with rapid frame rates. The basic spectral scanning architecture is demonstrated in Fig. 11(a). As the wavelength of a laser source varies over time, the output direction of the laser beam also changes correspondingly. A laser source works as a spectral-temporal modulator, while diffractive optics establish a one-to-one mapping relationship between

spectral and spatial positions. This enables precise control over the beam direction. Such systems exhibit rapid optical frequency scanning speed. Spectral scanning LiDAR systems also offer an advantage in terms of anti-interference ability. As depicted in Fig. 11(a), a laser beam with a specific wavelength (red color) passes through the diffractive optics and then reaches the surface of the target. Scattering occurs on the surface of the targets and only the echo signal with the same wavelength (red) will be collected by the LiDAR system.

If another LiDAR system with a different wavelength (blue color) detects the same position, the echo signal (blue) will be filtered out by the diffractive optics, ensuring that only the echo signal corresponding to the correct angle and wavelength is collected. The scanning rate and FoV of spectral scanning methods largely depend on the characteristics of the laser source and the design of the diffractive optics. To achieve large scanning FoV, the wavelength scanning range of the laser should be as wide as possible. However, it can be challenging to obtain a widely tunable laser that covers a broad range of wavelength. In terms of diffractive optics, gratings are commonly used for beam deflection.

A. Pulsed LiDAR System Based on Spectral Scanning

The time-stretch technique has been proposed to provide a fast wavelength-swept source for a spectrally scanning LiDAR system [30]. The basic structure diagram is demonstrated in Fig. 11(b). A broadband source is employed to generate a wide spectral scanning range for a large scanning FoV. Time-stretch technology is one of the options. Different methods are

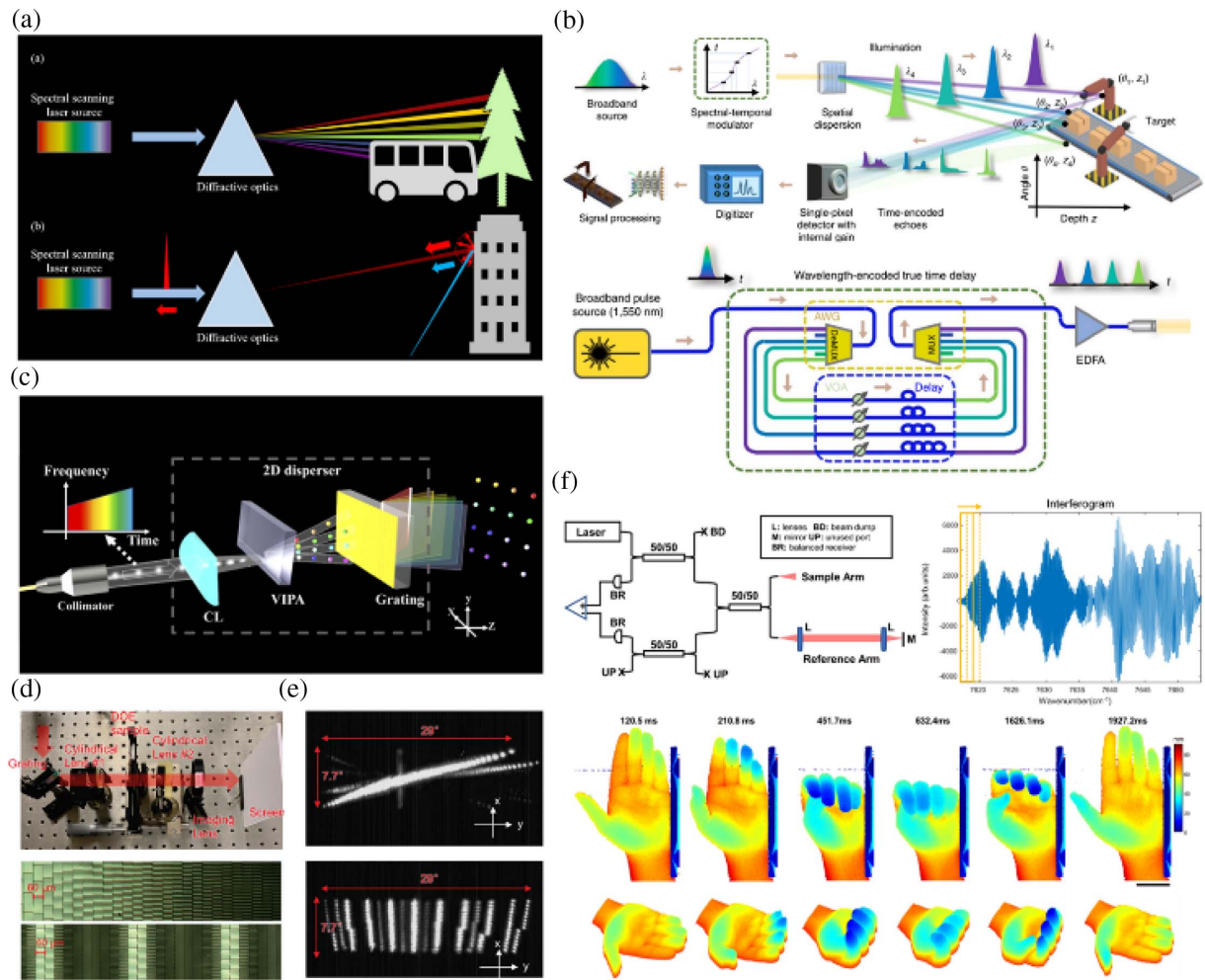


Fig. 11. (a) Illustration of the principle and anti-interference ability of spectral scanning LiDAR. (b) Basic architecture of spectral scanning pulsed LiDAR system. A spectro-temporal modulated laser source was used. (c) Virtual imaged phased array (VIPA)-based 2D disperser for 2D spectral scanning LiDAR. (d) Top: experimental setup of diffractive element for 2D spectral scanning. Bottom: microscopy image of two DOEs. (e) 1D (top) and 2D (bottom) spectral scanning performance. (f) Fast 3D imaging FMCW LiDAR system. Top: schematic diagram of swept-source FMCW LiDAR system. Short-time Fourier transform (STFT) is used for data processing. Bottom: video-rate 3D imaging results of human hands. (b) Reprinted with permission from Ref. [30], copyright 2020, Springer Nature. (c) Reprinted with permission from Ref. [125], copyright 2021, Optica Publishing Group. (d), (e) Reprinted with permission from Ref. [126], copyright 2021, Optica Publishing Group. (f) Reprinted with permission from Ref. [127], copyright 2022, Springer Nature.

available for constructing a time-stretch-based wavelength-swept laser source. As shown in Fig. 11(b), one method is introduced for pulsed LiDAR application and more implementations can be explored, as described in Ref. [128]. The time-stretch technique enables a line scanning rate of approximately 1 MHz, as demonstrated in Ref. [30]. It is important to note that time-stretch can only perform one-dimensional spectral scanning; thus, an additional beam-steering scanner is required to achieve 2D beam steering. When considering the spectral scanning rate alone, the point acquisition of time-stretch LiDAR can reach around 100 MHz. Implementation of such a LiDAR system can indeed achieve beam-steering speed limited by the time-of-flight limitation. Facing future high-frame-rate 3D imaging LiDAR, a time-stretch LiDAR system would serve as a valuable solution.

B. FMCW LiDAR System Based on Spectral Scanning

Pulsed LiDAR requires a higher bandwidth for electronic circuits, especially for ultra-fast 3D imaging LiDAR systems, as a large amount of data processing imposes a huge burden on the system. In comparison, FMCW LiDAR shows low requirements for electronic bandwidth, making it reliable for high-frame-rate LiDAR. In Refs. [127,129], an akinetic all-semiconductor swept source was used for fast-axis spectral scanning, enabling video-rate 3D imaging. The system incorporated the temporal-spectral multiplex technology for beam steering and ranging. The experimental setup is illustrated in Fig. 11(f). The proposed system could achieve beam steering and ranging simultaneously by employing short-time Fourier transform (STFT), as depicted in Fig. 11(f). This particular operating mode was also implemented in other LiDAR systems

[125,130]. The high performance of the system is attributed to the swept source, which achieves a maximum sweep rate of 200 kHz with 65.85-nm bandwidth. When operating at the maximum sweep rate, a potential point acquisition rate of 95 MHz can be realized. The experimental results demonstrated a 3D imaging frame rate of several tens of hertz.

However, a disadvantage of such LiDAR systems is the limited coherent length of the laser source, which restricts the maximum detectable ranging capability. In Refs. [127,129], only a tens of centimeters ranging experiment was conducted. As an alternative, a vertical-cavity surface-emitting laser (VCSEL) is possible for long-range and rapid 3D imaging FMCW LiDAR systems. With VCSELs, the coherent range of FMCW LiDAR can extend to several meters or even tens of meters [130]. The linewidth of a laser source could be further reduced by employing a mechanically steered semiconductor laser source, enabling coherent length of up to 100 m, but at the cost of low beam-steering speed [125]. In the current spectral scanning FMCW LiDAR system, realizing two-dimensional beam steering remains a challenge. Other beam-steering devices often result in slower scanning rates and increased system complexity. Therefore, it is crucial to explore diffractive optics for enabling 2D spectral scanning. Proposed solutions include VIPA-based 2D dispersers [131] and arrayed diffractive elements [126], as depicted in Figs. 11(c)–11(e). These diffractive-optics-based approaches aim to overcome the challenge of efficient and rapid 2D scanning while maintaining high scanning rates and system control simplicity.

5. PARALLEL DETECTION METHOD FOR HIGH-SPEED LIDAR

In the above sections, we discuss the limitations of serial detection for fast 3D imaging LiDAR systems. Regardless of the type of beam scanner used in the LiDAR system, the scanning speed is ultimately limited by the speed of light or ToF. To break such a physical limitation on the scanning speed, parallel detection has emerged as a powerful technology. As a direct solution to achieve parallelism, a pixelated detector array or multi-channel transceiver is the most adopted method. Current research in this area focuses on the design and fabrication of these devices to optimize their performance. The second method involves algorithm-related imaging techniques that use a single-pixel detector, such as compress sensing. However, the speed of these techniques is typically slower than that of serial scanning methods. The third method involves an optical coding method for parallel detection. In the following sections, we will provide a detailed discussion of these approaches and their performance in terms of speed.

A. Multiple Detectors for Parallel Detection

Flash LiDAR. By using an array of photodiodes, flash LiDAR can perceive the surrounding world without the assistance of beam scanners. It behaves like a camera and an array of photodiodes captures the echo signal from the scene. To illuminate the entire scene, expanding optics or diffusers are used in the transmitter to separate a single laser beam. The basic block diagram of flash LiDAR is depicted in Fig. 2(b). The detector array is usually integrated with CMOS timing circuitry for range measurement.

The optical part of the system incurs little time consumption. With just one round-trip time, the echo signal from the entire target can be received by the detector. The electric circuit should have a high bandwidth to accommodate the analog-to-digital converter and ToF measurement. The receiving FoV of the detector array should match the illuminating FoV to ensure accurate spatial mapping. In flash LiDAR, each pixel in the detector array has a one-to-one mapping with a spatial position, thereby determining the system's spatial resolution. The separation of the output laser beam leads to a reduced intensity, which hinders long-distance detection capabilities.

For flash LiDAR with the direct time-of-flight (dToF) ranging method, a single-photon avalanche diode (SPAD) array is the commonly used detector because of the extremely high sensitivity [132]. Another pixelated detector formed by numerous SPADs, i.e., multi-pixel photon counters (MPPCs) or named silicon photomultipliers (SiPMs), is also adopted for dToF measurement. A dToF flash LiDAR system requires low complexity of optical devices, offering a small-sized and low-cost solution for 3D imaging. The narrow pulse width ensures high peak optical power as well as eye-safety consideration. The round-trip delay is measured by high-resolution time-to-digital converters (TDCs) integrated on one single chip with SPADs, benefiting from the CMOS technology [133]. Thus, the dToF flash LiDAR can be used in outdoor applications ranging from tens to hundreds of meters. In Ref. [134], a 252×144 SPAD pixel sensor named Ocelot is presented. Every 126 pixels in a half-column of this sensor share six 12-bit TDCs with 48.8-ps resolution. In the time-correlated single-photon counting (TCSPC) mode, a dToF flash LiDAR based on this sensor achieves 3D imaging at 0.7 m with 30 frames per second (fps). Similarly, another study [31] presents a 256×256 SPAD sensor with a high-resolution TDC for high-speed 3D ToF applications, as shown in Fig. 12(a). The LiDAR measurements show the 64×64 image resolution with a frame rate of 30 fps. In Ref. [139], a 64×64 pixel SPAD sensor operating in both dToF mode and photon counting mode is reported. The sensor adopts a timing technique with a TDC and a time-to-amplitude converter (TAC), achieving a depth resolution of 1.95 mm and a maximum range of 32 km. The LiDAR system based on this sensor offers an FoV of $8.7^\circ \times 7.4^\circ$ with 7.3-mm accuracy within a range of 50 m.

Flash LiDAR can also be implemented based on the iToF ranging method. In this case, ToF cameras manufactured by standard CMOS technology are popular detector arrays. These small and inexpensive detectors provide another compact solution for 3D imaging. The crucial aspect of this approach lies in the different ways of modulating the light intensity to solve the ambiguous range problem [140]. The iToF CMOS image sensors commonly operate in the near-infrared (NIR) region with an integrated amplifier and high-resolution analog-to-digital converter (ADC) for fast signal readout [141–143]. Because of the limited optical power and the vulnerability to the background light, the iToF flash LiDAR is restricted to indoor applications within several to tens of meters [132]. The state-of-the-art dToF and iToF flash LiDAR sensors as well as the ToF cameras [144–146] available on the market are summarized in Table 4.

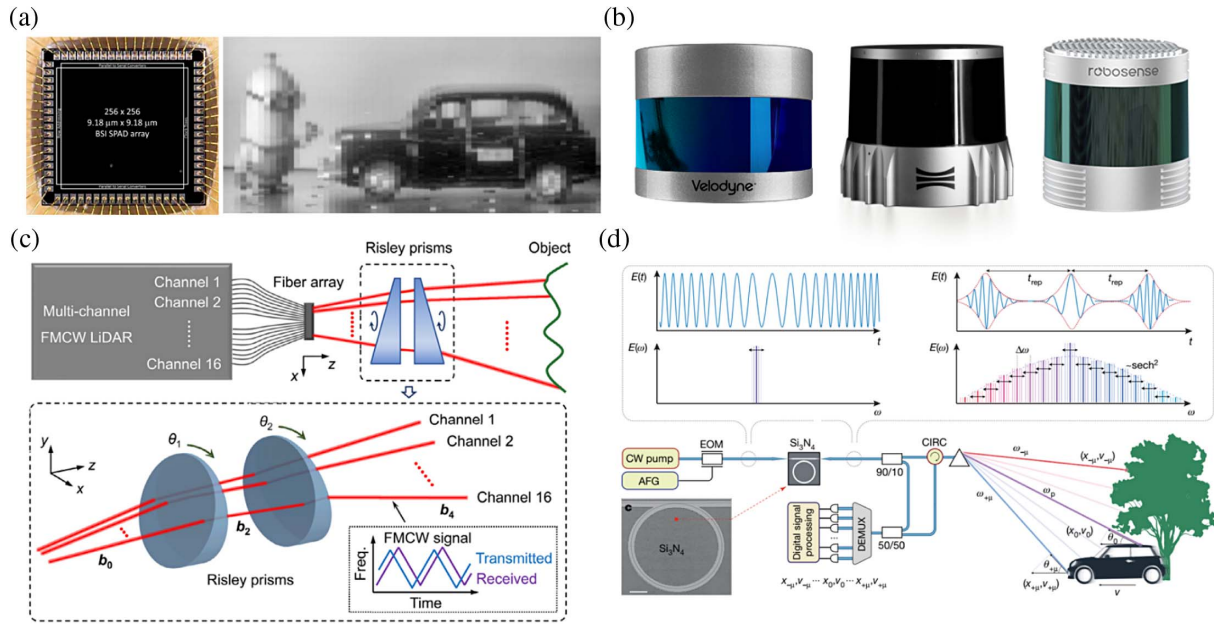


Fig. 12. (a) SPAD sensor for flash LiDAR and the imaging results. (b) Three types of multiple-channel scanning LiDAR systems from commercial products, Velodyne LiDAR (Ultra Puck), Ouster (OS2), and Robosense (RS-Ruby). (c) Structure of proposed Risley-prism-based multi-beam scanning LiDAR. Experimental setup and working principle of multiple-channel scanning. (d) Working principle of parallel FMCW LiDAR using soliton microcomb. (a) Reprinted with permission from Ref. [31], copyright 2019, IEEE. (b) Provided by Refs. [135–137]. (c) Reprinted with permission from Ref. [138], copyright 2022, SPIE. (d) Reprinted with permission from Ref. [32], copyright 2020, Springer Nature.

Table 3. Comparison of LiDAR Performance for Selected Solid-State Beam Scanners

Ref. and Type	1D or 2D	Solid-State Scanner	Steering Angle (°)	Phase Tuning Mechanism	Angular Resolution (°)	Element Number	Point Acquisition Rate (kpixels/second)
[90] Research	2D	Optical phased array	23×3.6	Thermal-optic	1.3	23×6	~ 10
[91] Research	1D	MEMS phased array	87	MEMS actuated	1.3×10^{-3}	670	350
[93] Research	1D	Optical phased array	45.4	Electronic-optic	—	16 (1×16)	20×10^3
[94] Research	1D	Optical phased array	6	Multiple quantum well	—	—	18×10^6
[29] Research	2D	Focal plane switch array	70×70	MEMS actuated	0.6	128×128	$\sim 10^3$
[103] Research	2D	Focal plane switch array	1×1	MEMS actuated	0.1	100 (10×10)	100
[97] Research	2D	Focal plane switch array	1.5×0.9	Thermal-optic	—	512 (32×16)	5×10^3
[105] Research	2D	Slow-light grating	5.1×2.8	Thermal-optic	—	4928 (154×32)	$\sim 10^3$
[25] Research	2D	Metasurface	6×4	Electronic-optic	—	651 (31×21)	5.4×10^3
[117] Research	2D	DMD-based beam scanner	12.6	—	0.97	—	23
[120] Research	1D	Acousto-optic beam scanner	18	Acousto-optic	0.12	—	670

Hybrid LiDAR. Combining multiple detector arrays with scanning methods, high-performance LiDAR systems capable of long detectable distances, large FoVs, and high point acquisition rates can be realized. This approach is widely adopted by many commercial LiDAR companies. Here, multiple laser diodes are commonly used to transmit detection beams, which is so-called multiple-channel detection. For instance, Velodyne's LiDAR system utilizes 32 laser sources for multiple-channel detection, achieving a data acquisition rate of 695,000 points per second [135]. Moreover, Ouster [136] and Robosense [137] have developed parallel detection LiDAR systems with

64 or 128 channels, achieving maximum point acquisition rates of 2.62 MHz and 2.3 MHz, respectively. However, in such multiple-channel LiDAR systems, the rotating head is typically used for beam steering for all channels. Three examples of such LiDAR systems are shown in Fig. 12(b).

Except for the spinning head for multiple-channel parallel detection, Risley-prism-based scanners are also a vital tool for parallel detection [136]. The basic concept is demonstrated in Fig. 12(c), where Risley prisms steer multiple beams simultaneously, allowing each beam to scan over a distinct area. In Ref. [138], FMCW ranging technology is used for ranging

Table 4. State-of-the-Art iToF Flash LiDAR Sensor and ToF Cameras Available on the Market

Ref. and Type	Detection Method	Pixel Array	FoV (°)	Maximum Detection	Frame	Point Acquisition Rate (kpixels/second)
				Range (m)	Rate (fps)	
[31] Research	dToF	64 × 64	1.2 × 1.2	50	30	122
[134] Research	dToF	252 × 144	40 × 20	50	30	1088
[139] Research	dToF	64 × 64	8.7 × 7.4	32,000	8300	33,997
[141] Research	iToF	1024 × 1024	120 × 120	4.2	30	31,457
[142] Research	iToF	640 × 480	25 × 19	30	120	36,864
[143] Research	iToF	1280 × 960	78	4	60	73,728
[144] Commercial	iToF	512 × 424	70 × 60	4.2	60	13,025
[145] Commercial	iToF	220 × 128	120 × 70	15	10	282
[146] Commercial	iToF	640 × 480	59 × 45	6	30	9216

operations. This approach enables the generation of point clouds with multiple channels, providing variable and flexible scanning density. Commercial LiDAR companies have embraced Risley-prism-based multiple-channel LiDAR systems. One prominent example is the Livox series developed by DJI company [147], which features a 64-channel scanning LiDAR with a data acquisition rate of around 300,000 points per second.

The combination of the spectral scanning method with the multiple-channel working mechanism in LiDAR systems can significantly enhance the data acquisition rate. Riemsberger *et al.* proposed a massively parallel coherent LiDAR system using a soliton microcomb as the laser source [32]. The basic working principle is demonstrated in Fig. 12(d). In their experiment, 30 distinct channels were utilized for parallel detection, resulting in an impressive point acquisition rate of megapixels per second. The limitations of their system, including the limited spectral channels and the requirement of massive photodiodes, prompted the development of a hardware-efficient method by the same research group. The optimized approach involves the use of two triangular chirped soliton microcombs and a single photodiode, enabling 64-channel distance and velocity measurement [148]. By leveraging the dual-comb technique, this LiDAR system achieves an impressive acquisition rate of 5 megapixels per second.

B. Single-Pixel Detectors for Parallel Detection

The single-pixel detectors for multiple-channel parallel detection can significantly increase the cost of LiDAR systems. It is crucial to explore the implementation of parallel detection in single-pixel LiDAR systems. In conventional pulse-based LiDAR systems, it is challenging to distinguish the echo signals received from different positions of the target if they arrive at the receiver simultaneously. However, by employing the direct-sequence optical code division multiple access (DS-OCDMA) technique [149], it becomes possible to mark each detecting beam with specific information to identify its illumination position. This eliminates the need to consider the sequence of transmitted pulses and allows for parallel detection. The depth information can be calculated by demodulating the signal at the receiver end. Fersch *et al.* conducted experiments to verify this technique [150]. Yang *et al.* also utilized an APD array and a rotating encoder plate to achieve OCDMA parallel 3D imaging LiDAR [33], with a point acquisition rate of 4 MHz [151]. Compressed sensing is a technique that allows for the reduced data acquisition without degrading the imaging quality, thereby saving time. Ye *et al.* demonstrated the compressed sensing method with a modified forward model to reduce the number of scanning points and capture time. The proposed architecture in Fig. 13(a), using the SPAD, was capable of imaging scenes

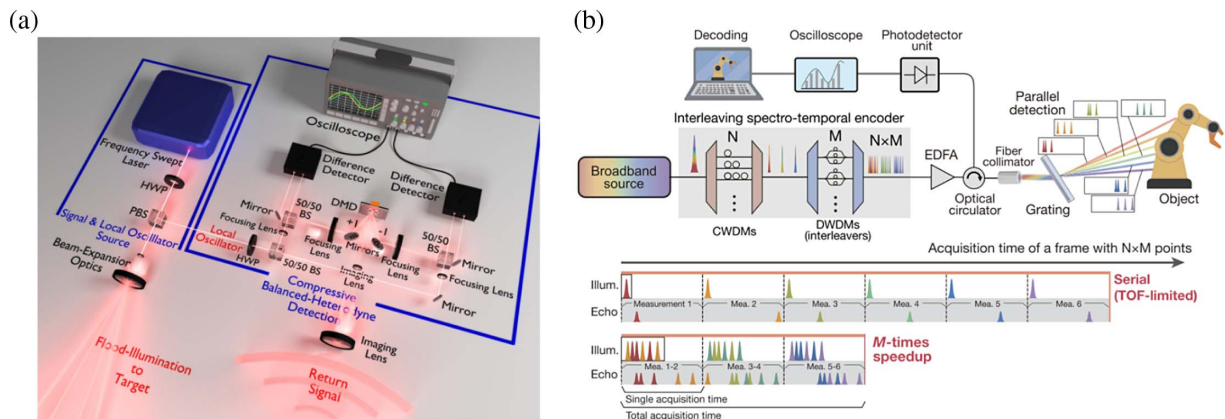


Fig. 13. (a) System architecture of compressive FMCW-LiDAR depth mapping. (b) Top: system structure of all-optical spectro-temporal encoding LiDAR system. Middle: illumination and echo signal of a serial LiDAR. Bottom: correlated spectro-temporal encoding enables parallelism and speeds up the LiDAR M -times beyond the time-of-flight limit. (a) Reprinted with permission from Ref. [34], copyright 2018, Optica Publishing Group. (b) Reprinted with permission from Ref. [152], copyright 2022, American Institute of Physics.

at a spatial resolution of 64×64 pixels through only 5×5 scanning points [34]. This approach enables faster data acquisition and reduces the computational burden without sacrificing the quality of the resulting images.

We recently combined spectral scanning with OCDMA to realize an ultra-fast 3D imaging LiDAR system [152]. The all-optical interleaved spectro-temporal encoder [Fig. 13(b)] determines the total ranging points based on $N \times M$ spectral channels, where M types of codewords enable parallel detection in M channels. The use of coarse and dense wavelength division multiplexers (CWDMs and DWDMs) allows efficient separation of spectral channels. The free spectral range of DWDM can be exploited as an interleaving spectral encoder. Time-domain decoding extracts the flight time of each channel by correlating the sampled echo signals with a pre-sampled code word template. Experimental results demonstrate impressive performance, achieving 4.41 and 3.56 times the ToF-limited speed for ranging distances of 75 m and 25 m, respectively, corresponding to record acquisition rates of 8.82 MHz and 21.38 MHz. This innovative parallelism approach is expected to open new avenues for the advancement of ultrafast single-pixel LiDAR systems.

6. SUMMARY AND OUTLOOK

In this paper, we analyze the basic working mechanism of LiDAR systems and initiate a fundamental discussion on the key factors that limit imaging speed. Serial detection and ToF methods physically constrain the maximum point acquisition rate of LiDAR systems. Various ranging techniques yield different achievable speeds, revealing their distinct performance, as illustrated in Table 5.

Among these techniques, pulsed LiDAR shows the potential to achieve imaging speeds approaching the ToF limitation. In current LiDAR systems, the main factor hindering speed improvement, excluding electronic signal processing time on circuits, is the beam-steering speed of beam scanners. We have analyzed the beam-steering performance of different kinds of scanners, beginning with traditional mechanical beam scanners. These scanners rely on the motor operating speed and generally

have a low point acquisition rate ranging from 10 kHz to 100 kHz.

In contrast, solid-state beam scanners, such as OPA and FPSA, provide fast and stable beam-steering capabilities, with point acquisition rates reaching MHz levels. Moreover, these solid-state beam scanners can be integrated on a chip, enhancing their compactness and versatility.

Spectral scanning LiDAR, an innovative scanning technology, stands out in terms of beam scanning rate. It benefits from ultrafast light sources, enabling high-speed scanning. Time-stretched dispersive scanning light sources, optical frequency combs, and tunable semiconductor lasers are representative examples. By incorporating dispersive optical components, the system can achieve a several-tens MHz data acquisition rate, approaching the fastest point acquisition rate but only limited by the theoretical ToF of light.

However, this area still requires further investigation. The first challenge is improving the FoV of spectral scanning, which necessitates enhancements in the dispersion performance of the diffractive optics and the wavelength-swept range of the light source. Another significant challenge is achieving 2D beam steering without relying on a combination of spectral scanning and mechanical scanners. The current approach involves separately scanning the fast and slow axes. However, achieving higher speeds requires devices capable of 2D spectral scanning, placing stringent demands on the 2D disperser. This includes achieving a large 2D scanning FoV, up to 100° , and spectral scanning within a limited tuning bandwidth, requiring large dispersion coefficients for dispersive devices. Integration and miniaturization of the technology are also vital aspects to be considered. Grating couplers serve as a powerful tool for implementing chip-scale spectral scanning but need to be combined with on-chip beam-steering methods such as FPSA or OPA. Meanwhile, the characteristics of waveguides, such as the spectral bandwidth, should be compatible with ultra-broadband spectrum sources.

Parallel detection is indispensable to overcome the inherent speed limitations of serial detection methods based on ToF. Commercial LiDAR systems currently achieve video-streaming

Table 5. Summary of Several LiDAR Imaging Methods

Sensing Mechanism		Detection Range	Axial Resolution	Angular Resolution	FoV	Point Acquisition Rate	Cost	Integration Level
Mechanical beam scanners	Polygonal mirror	—	—	Medium	Large	Medium	Low	Low
	Galvanometric mirrors	—	—	Medium	Small	Low	Low	Low
	Risley prism	—	—	Low	Large	Low	Medium	Low
	MEMS mirrors	—	—	High	Large	High	High	Medium
Solid-state beam scanners	Optical phased array	—	Low	Low	Small	High	High	High
	Focal plane switch array	—	Medium	High	Large	Medium	High	High
	Slow-light grating	—	—	Medium	Small	High	High	High
	Metasurface	—	—	High	Small	High	High	High
Parallel detection	Spectral scan	—	High	High	Small	High	Low	Medium
	Flash LiDAR	Short	—	Medium	Medium	Medium	High	Medium
	Hybrid LiDAR	Long	Medium	Low	Large	Medium	Low	Medium
	Micro-comb	—	High	High	Small	High	High	High
	Optical coding	Long	—	High	Small	High	High	Medium

imaging by multiple detection light sources and detectors, reducing the burden on the beam scanning device to meet imaging resolution requirements. However, the use of multiple probes increases the cost and system sizes. Therefore, a single-pixel parallel detection LiDAR scheme based on optical coding represents a competitive solution. Optical orthogonal codes enable simultaneous detection of multiple channels, and decoding at the receiving end extracts the ranging information from different channels. While optical coding adds complexity to the system, combining spectral scanning and optical encoding promises the fastest single-pixel imaging LiDAR system, as confirmed by experiments.

In fact, several challenges persist, such as improving the signal-to-noise ratio of decoding and achieving wide-spectrum signal encoding. In general, ultrafast LiDAR is still in the development stage. As the LiDAR market continues to evolve, this area will undoubtedly attract more attention and investment, leading to advancements and breakthroughs in achieving higher imaging speeds and improved LiDAR performance.

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